

PYTHON LIST PROGRAMS WITH OUTPUT

1. Empty List

PROGRAM:
empty list
my_list = []
print(my_list)

OUTPUT:

[]

2. List of Integers

PROGRAM:
list of integers
my_list = [1, 2, 3]
print(my_list)

OUTPUT:

[1, 2, 3]

3. List with Mixed Data Types

PROGRAM:
list with mixed data types
my_list = [1, "Hello", 3.4]
print(my_list)

OUTPUT:

[1, 'Hello', 3.4]

4. Multiple Data Type Assignment

PROGRAM:
multiple datatype assign
n_list = [2.9, 3000, 'Hello', "Welcome to python list ", True]
print(n_list)

OUTPUT:

[2.9, 3000, 'Hello', 'Welcome to python list ', True]

5. Nested List Example 1

PROGRAM:
n_list = ["Happy", [2, 0, 1, 5]]
Nested indexing
print(n_list[0][3])
print(n_list[1][2])

OUTPUT:

p
1

6. Nested List Example 2

PROGRAM:

```
my_list = ["mouse", [8, 4, 6], ['a', 'b', 'c']]
print(my_list)
print(my_list[0])
# Nested indexing
print(my_list[1][1])
print(my_list[2][2])
```

OUTPUT:

```
['mouse', [8, 4, 6], ['a', 'b', 'c']]
mouse
4
c
```

7. Negative Indexing in Lists

PROGRAM:

```
my_list = ['p', 'r', 'o', 'b', 'e']
print(my_list[-1])
print(my_list[-5])
```

OUTPUT:

e
p

8. Appending and Extending Lists

PROGRAM:

```
odd = [1, 3, 5]
odd.append(7)
print(odd)

odd.extend([9, 11, 13])
print(odd)
```

OUTPUT:

```
[1, 3, 5, 7]
[1, 3, 5, 7, 9, 11, 13]
```

9. Concatenating and Repeating Lists

PROGRAM:

```
odd = [1, 3, 5]

print(odd + [9, 7, 5])
```

```
print(["re"] * 3)
```

OUTPUT:

```
[1, 3, 5, 9, 7, 5]
['re', 're', 're']
```

10. Length of a List

PROGRAM:

```
thislist = ["apple", "banana", "cherry"]
print(len(thislist))
```

OUTPUT:

```
3
```

11. Remove Specified Item

PROGRAM:

```
thislist = ["apple", "banana", "cherry"]
thislist.remove("banana")
print(thislist)
```

OUTPUT:

```
['apple', 'cherry']
```

12. Sort a List of Strings

PROGRAM:

```
thislist = ["orange", "mango", "kiwi", "pineapple", "banana"]
thislist.sort()
print(thislist)
```

OUTPUT:

```
['banana', 'kiwi', 'mango', 'orange', 'pineapple']
```

13. Sort a List of Numbers

PROGRAM:

```
thislist = [100, 50, 65, 82, 23]
thislist.sort()
print(thislist)
```

OUTPUT:

```
[23, 50, 65, 82, 100]
```

14. Insert an Item into a List

PROGRAM:

```
fruits = ['apple', 'banana', 'cherry']
fruits.insert(1, "orange")
print(fruits)
```

OUTPUT:

```
['apple', 'orange', 'banana', 'cherry']
```

15. Count an Item in a List (String)

PROGRAM:

```
fruits = ["apple", "banana", "cherry", "apple"]
```

```
x = fruits.count("apple")  
print(x)
```

OUTPUT:

```
2
```

16. Count an Item in a List (Number)

PROGRAM:

```
fruits = [1, 4, 2, 9, 7, 8, 9, 3, 1]  
x = fruits.count(9)  
print(x)
```

OUTPUT:

```
2
```

17. Find Index of a Value

PROGRAM:

```
lst = [10, 20, 30, 40, 50]
```

```
value = 30
```

```
position = lst.index(value)
```

```
print("List      :", lst)  
print("Value     :", value)  
print("Index     :", position)
```

OUTPUT:

```
List      : [10, 20, 30, 40, 50]  
Value     : 30  
Index     : 2
```